A Geometrical Synthesis of Maxwellian Electrodynamics: A Guide for the Modern Electrical Engineer

The pedagogical architecture of undergraduate electrical engineering has long been dominated by the vector calculus of Oliver Heaviside and Josiah Willard Gibbs. While this framework provides a robust computational toolkit for the design of circuits and antennas, it frequently obscures the profound geometric relationships that underpin electromagnetic theory. The reliance on the nabla operator (∇) for gradient, divergence, and curl often leads to a "diabolic" mechanical approach to physics, where students manipulate symbols without a clear visual intuition of the fields themselves.[1] In recent years, the revival of Élie Cartan's calculus of differential forms—specifically through the lens of Tristan Needham's "Visual Differential Geometry and Forms"—has offered a transformative path forward. By treating fields as geometric "atoms" such as surfaces and tubes, the engineer can perceive Maxwell's equations as fundamental statements about the topological and metric structure of space and time.[1]

The Newtonian Foundation: Ultimate Equality and Geometric Reasoning

A central tenet of the Needham approach to differential geometry is the restoration of Newton's "geometrical methods" to the center of mathematical analysis. For the electrical engineering student, this begins with the concept of "Ultimate Equality," denoted by the symbol $\asymp$.[2] In traditional calculus, the concept of a limit is often handled with ϵ-δ rigor, which, while mathematically sound, can feel disconnected from the practical realities of approximating field behavior in a finite-element grid or at the edge of a conductor.

Ultimate Equality posits that two quantities A and B, which depend on a small, vanishing parameter ϵ, are "ultimately equal" if their ratio tends to unity as ϵ approaches zero.[2] Mathematically:

$$A \asymp B \iff \lim_{\epsilon \to 0} \frac{A}{B} = 1$$

This perspective allows the engineer to treat infinitesimal elements of space and time as concrete geometric objects. When calculating the electromotive force around a small loop or

the flux through a surface, one does not need to get lost in the machinery of limits; rather, one identifies the geometric relationships that hold "in the limit".[2] This philosophy mirrors the discrete nature of modern computational electromagnetics, where fields are evaluated on a grid and the local linearity of the field is the primary assumption for differentiability.[9]

The Geodesic as a Straight Edge

Needham further emphasizes the intrinsic nature of geometry by defining a straight line, or a geodesic, as the shortest distance between two points on a curved surface.[6] For an inhabitant of a curved manifold, such as a "Spherelander" or a "Hyperbolic-Spherelander," the local physics is determined entirely by these geodesics. A practical demonstration involves peeling a thin strip of rind from a fruit, such as a squash or pomelo, and laying it flat on a table. If the strip was cut along a geodesic, it becomes a straight line in the Euclidean plane.[6] This intrinsic perspective is vital for electromagnetics, as it prepares the student to understand that fields are not merely "arrows" in an empty void but are tied to the metric properties of the space they inhabit—a realization that becomes critical when moving from classical electrodynamics to general relativity.[1]

The Taxonomy of Differential Forms in Three Dimensions

To construct a guide to Maxwell's equations using differential forms, one must first categorize the physical quantities involved based on their geometric behavior. Vector analysis fails to distinguish between field intensities (like $\mathbf{E}$ and $\mathbf{H}$) and flux densities (like $\mathbf{D}$ and $\mathbf{B}$), representing both as vectors.[4] Differential forms resolve this by assigning each quantity to a form of degree k, which dictates how it is integrated and visualized.

0-Forms: The Scalar Potential as Point Assignments

At the base of the hierarchy are 0-forms, which are simply smooth functions $f(x, y, z)$ on a manifold.[15] In electromagnetics, the scalar potential Φ is a 0-form. Geometrically, a 0-form is "integrated" over a 0-dimensional object—a point. Evaluating a function at a point P is denoted as $\int_P f = f(P)$.[15] If the point is oriented negatively (e.g., Q), the evaluation is $\int_Q f = -f(Q)$. This establishes the fundamental principle of orientation that carries through all higher-order forms.[15]

1-Forms: Field Intensities as Surfaces

A 1-form is the natural language for field intensities such as the electric field $\mathbf{E}$ and the magnetic field $\mathbf{H}$.[4] While vector calculus represents $\mathbf{E}$ as a vector pointing in the direction of the force on a test charge, the differential form approach represents $\mathbf{E}$ as a collection of oriented surfaces.[14]

Geometrically, a 1-form $\omega = adx + bdy + cdz$ defines a family of surfaces at every point. The "magnitude" of the 1-form is represented by the density of these surfaces. When an engineer integrates a 1-form along a path $\mathcal{C}$, the result $\int_{\mathcal{C}} \omega$ is the number of surfaces pierced by the path.[4] In the case of the electric field $\mathbf{E}$, these surfaces are levels of constant potential. Piercing three surfaces of $\mathbf{E}$ along a path signifies a potential difference of three units. This visualization clarifies why $\mathbf{E}$ is an intensity: it measures the "effort" or "density" of potential change across a distance.[4]

Coordinate	Basis 1-Form	Visual Representation
x	dx	Planes perpendicular to the x-axis.
y	dy	Planes perpendicular to the y-axis.
z	dz	Planes perpendicular to the z-axis.
r	dr	Concentric cylinders or spheres.

2-Forms: Flux Densities as Tubes

Flux quantities, including electric displacement $\mathbf{D}$, magnetic flux density $\mathbf{B}$, and current density $\mathbf{J}$, are represented as 2-forms.[4] Whereas a 1-form is a stack of surfaces, a 2-form is a "honeycomb" of tubes.[4]

The integral of a 2-form over a surface $\mathcal{S}$ counts the number of tubes that pass through that

surface. For the magnetic flux density $\mathbf{B}$, this integral yields the total magnetic flux Φ_B.[8] The use of 2-forms provides a direct visual link to the concept of "lines of force" popularized by Faraday, but with a more rigorous mathematical backing: the "lines" are actually tubes with finite cross-sectional area determined by the field strength.[4]

The basis for 2-forms in three dimensions is formed by the wedge products of basis 1-forms: $dy \wedge dz$, $dz \wedge dx$, and $dx \wedge dy$.[16] Each basis 2-form represents a family of tubes aligned with one of the coordinate axes. For example, $dx \wedge dy$ represents tubes pointing in the z-direction, formed by the intersection of dx-planes and dy-planes.[16]

3-Forms: Sources as Cells

The charge density ρ is a 3-form. Geometrically, a 3-form is a collection of infinitesimal "cells" or "boxes".[4] Integrating a 3-form over a volume $\mathcal{V}$ yields a scalar value—the total charge Q.[4] In the differential form framework, the 3-form $\rho dx \wedge dy \wedge dz$ is an entity that exists only as a volume density, ensuring that the dimensions of the integrand and the domain of integration are always consistent.[15]

Exterior Algebra: The Mechanics of Interaction

To describe how fields interact, we must employ the wedge product ($\wedge$), which is the fundamental operation of exterior algebra.[16] The wedge product is anti-commutative, meaning that the order of multiplication determines the orientation.[16]

The Geometric Meaning of the Wedge Product

The wedge product of two 1-forms $\alpha \wedge \beta$ creates a 2-form. Visually, this is the intersection of two families of surfaces.[14] If α is a set of horizontal planes and β is a set of vertical planes, their wedge product $\alpha \wedge \beta$ is the set of horizontal tubes formed by their crossing.[14]

This has a critical application in the Poynting vector, which in vector calculus is the cross product $\mathbf{S} = \mathbf{E} \times \mathbf{H}$. In differential forms, the Poynting power flow is a 2-form:

$$S = E \wedge H$$

The surfaces of the electric 1-form E and the magnetic 1-form H form the "sides" of the tubes of the Poynting power flow 2-form S. This provides an immediate, intuitive explanation for why power flow is orthogonal to both E and H: the intersection of two families of surfaces naturally defines a direction (along the tubes) that is perpendicular to the normals of both original families of surfaces.[4]

Form Interaction	Resulting Degree	Geometric Interpretation
1-form $\wedge$ 1-form	2-form (Tubes)	Intersection of two sets of surfaces.
1-form $\wedge$ 2-form	3-form (Cells)	Slicing tubes into boxes.
2-form $\wedge$ 1-form	3-form (Cells)	Slicing tubes into boxes.
k-form $\wedge$ k-form	0 (if identical)	Anti-commutativity ($dx \wedge$).

The Exterior Derivative: A Unified Operator

The single most powerful tool in this framework is the exterior derivative (d), which generalizes the gradient, curl, and divergence of vector calculus into a single operation.[5] The exterior derivative maps a k-form to a $(k+1)$-form. Its action is essentially to find the "boundary" or "net change" of a form across an infinitesimal region.[15]

One of the most profound identities in this calculus is $d^2 = 0$, meaning that the exterior derivative of an exterior derivative is always zero.[9] This single identity encompasses both $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ and $\nabla \times (\nabla f) = 0$. For the undergraduate student, this reduces the list of identities to memorize and replaces them with a topological truth: a boundary has no boundary.[9]

Mapping Maxwell's Equations to Differential Forms

Using these definitions, the four Maxwell equations can be rewritten in a way that reveals their

distinct topological and physical origins.[13]

The Topological Pair ($dF = 0$)

The first two Maxwell equations—Gauss's Law for Magnetism and Faraday's Law—describe the topological constraints on the fields. They state that the Faraday field tensor F is closed.[18]

1. **Gauss's Law for Magnetism**: $d\mathbf{B} =$. Visually, this means the tubes of the magnetic 2-form $\mathbf{B}$ never end. They form closed loops or extend to infinity.[4] There are no 3-form "boxes" of magnetic charge to serve as sources or sinks for these tubes.[4]
2. **Faraday's Law**: $d\mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$. This equation states that the 2-form "curl" of the electric field intensity ($d\mathbf{E}$) is equal to the negative time rate of change of the magnetic flux 2-form ($\mathbf{B}$). In the visual framework, as magnetic tubes are added to or removed from a region, they "push" against the surfaces of the electric 1-form, creating an induced intensity.[4]

The Source Pair ($dG = J$)

The second two equations—Gauss's Law and the Ampère-Maxwell Law—describe how matter (charges and currents) generates fields.[18]

1. **Gauss's Law**: $d\mathbf{D} =$. The exterior derivative of the 2-form displacement flux $\mathbf{D}$ is the 3-form charge density ρ. Geometrically, this is an exact statement of source and flux: the "cells" of charge are the end-points (the "caps") of the tubes of displacement flux. To find the total charge in a volume, one simply counts the number of displacement tubes crossing its boundary.[4]
2. **Ampère-Maxwell Law**: $d\mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$. This equation states that tubes of current ($\mathbf{J}$) and time-varying displacement flux ($\dot{\mathbf{D}}$) generate surfaces of magnetic field intensity ($\mathbf{H}$). Graphically, current tubes act as the "axis" around which magnetic surfaces wrap.[4]

Maxwell's Equation	Standard Vector Form	Differential Form	Unified Meaning
Gauss (Magnetism)	$\nabla \cdot \mathbf{B} = 0$	$d\mathbf{B} =$	Magnetic tubes have no end-points.

Faraday	$\nabla \times \mathbf{E} =$	$d\mathbf{E} = -\dot{\mathbf{B}}$	Changing B-flux generates E-surfaces.
Gauss (Electricity)	$\nabla \cdot$	$d\mathbf{D} =$	Charge cells emit D-flux tubes.
Ampère-Maxwell	$\nabla \times \mathbf{H} = \mathbf{J} +$	$d\mathbf{H} = \mathbf{J} +$	Current/D-flux tubes generate H-surfaces.

The Hodge Star and the Metric Structure of Space

While the equations $dF = 0$ and $dG = J$ describe the topology of the fields, they do not tell us how E relates to D or H relates to B. In vector calculus, these are simply related by constants ϵ and μ.[21] However, differential forms distinguish between the topological laws (which are coordinate-independent and hold even in curved space) and the metric laws (which depend on the local distance and angle measurements).[9]

The operator that bridges this gap is the Hodge star ($*$).[18] The Hodge star is a metric-dependent operator that maps a k-form to an $(n-k)$-form. In three dimensions, it maps a 1-form (surfaces) to a 2-form (tubes).[18]

Constitutive Relations and the Metric

For a linear, isotropic medium, the constitutive relations are expressed as:

$$\mathbf{D} = \epsilon * \mathbf{E}$$

$$\mathbf{B} = \mu * \mathbf{H}$$

The Hodge star takes the surfaces of the intensity 1-forms (E and H) and "rotates" them into the tubes of the flux density 2-forms (D and B). The density of the surfaces is converted into

the density of the tubes.[14]

This separation is of immense practical value for the engineer. It clarifies that the four Maxwell equations are actually two sets of universal topological laws, and the specific behavior of a material (whether it is a vacuum, a dielectric, or a plasma) is entirely encapsulated in the Hodge star.[4] This distinction is often lost in vector calculus, where ϵ and μ are treated as mere scaling factors.[5]

Practical Engineering Examples via Forms

To demonstrate the "concrete and practical" utility of this approach, we consider several foundational problems in electrical engineering.

The Parallel-Plate Capacitor: Energy in Boxes

Consider a parallel-plate capacitor with a voltage V and a charge Q. The electric field intensity $\mathbf{E}$ is a 1-form whose surfaces are parallel to the plates. The displacement flux $\mathbf{D}$ is a 2-form whose tubes extend vertically from one plate to the other.[4]

The energy density w is the 3-form:

$$w = \frac{1}{2} E \wedge D$$

Geometrically, the surfaces of E slice the tubes of D into infinitesimal "boxes" or "cells" of energy.[4] The total energy stored in the capacitor is the sum of these boxes. This visualization provides a much clearer intuition of energy storage than the vector dot product $\mathbf{E} \cdot \mathbf{D}$, as it shows how the two fields physically intersect to fill the volume with energy.[4]

The Coaxial Cable: Poynting Flux Tubes

In a coaxial cable carrying a DC current I and at a voltage V, the fields are as follows:

- The electric 1-form E has radial surfaces.
- The magnetic 1-form H has longitudinal surfaces (planes containing the axis).
- The current 2-form J is a set of tubes inside the center conductor.[14]

The Poynting 2-form $S = E \wedge H$ consists of tubes that are parallel to the axis of the cable.[14] The intersection of the radial E-surfaces and the longitudinal H-surfaces creates tubes that point straight down the cable, representing the flow of power from the source to the load. This confirms visually that power is carried not in the wires, but in the fields between the conductors.[14]

Plane Waves: The Geometry of Propagation

In a propagating electromagnetic wave, the electric 1-form E and the magnetic 1-form H are mutually perpendicular. Their wedge product $S = E \wedge H$ creates tubes of power flow that point in the direction of propagation.[14]

Using the Hodge star, we can see the "exchange" of energy. Faraday's Law $d\mathbf{E} = -\dot{\mathbf{B}}$ and the Ampère-Maxwell Law $d\mathbf{H} = \dot{\mathbf{D}}$ show that as the electric surfaces "bend" (the curl), they generate magnetic tubes, and as magnetic surfaces bend, they generate electric tubes.[14] This "leap-frogging" of forms is the geometric mechanism of wave propagation in a vacuum.[29]

Problem Component	Differential Form Representation	Physical Significance
Capacitor Charge	$d\mathbf{D} =$ (3-form)	Charge cells source D-tubes.
Inductor Magnetic Flux	$\int_S \mathbf{B}$ (2-form)	Counting tubes passing through a coil.
Transmission Power	$E \wedge$ (2-form)	Intersection of field surfaces creates power tubes.
Energy Storage	$E \wedge$ (3-form)	Slicing tubes into energy cells.

The Unified Integral Theorems: The Generalized Stokes Theorem

For an undergraduate student, the most significant mathematical simplification provided by differential forms is the unification of the fundamental theorems of vector calculus. The Generalized Stokes Theorem states [5]:

$$\int_{\Omega} d\omega = \int_{\partial\Omega} \omega$$

This single formula replaces the Fundamental Theorem of Calculus (for 0-forms), the Kelvin-Stokes Theorem (for 1-forms), and the Gauss Divergence Theorem (for 2-forms).[9]

Visualizing the Boundary-Bulk Relationship

In the context of the Divergence Theorem, the "bulk" integral is the sum of charge cells ($\rho = d\mathbf{D}$) within a volume $\mathcal{V}$.[4] The "boundary" integral is the count of displacement flux tubes ($\mathbf{D}$) passing through the surface $\partial\mathcal{V}$.[4] The theorem states that every tube crossing the boundary must originate from a cell inside—a statement that is topologically obvious but algebraically complex in vector notation.[4]

Similarly, for Faraday's Law, the "bulk" integral is the sum of the "bending" of electric surfaces over a surface area $\mathcal{S}$ ($d\mathbf{E}$). The "boundary" integral is the total potential difference along the loop $\partial\mathcal{S}$ ($\oint \mathbf{E}$). The theorem shows that the net potential around the loop is exactly equal to the density of the "swirl" of field surfaces inside.[4]

Electrodynamics in Four-Dimensional Spacetime

The ultimate power of differential forms is realized when time is treated as a coordinate $x^0 = t$, forming a four-dimensional Minkowski spacetime.[13] In this framework, the electric and magnetic fields are no longer separate entities but components of a single 2-form F, known as the Faraday field tensor.[18]

The Faraday 2-Form F

The 2-form F in 4D combines the 3D electric intensity 1-form and magnetic flux 2-form [18]:

$$F = E \wedge dt + B$$

The entirety of electromagnetism is then contained in just two equations [18]:

1. **The Homogeneous Equation**: $dF = 0$. This single equation combines Gauss's Law for Magnetism and Faraday's Law. It states that the field F is a "closed" object in spacetime.[18]
2. **The Inhomogeneous Equation**: $d * F = J$. This combines Gauss's Law (Electricity) and the Ampère-Maxwell Law. It relates the field to the source 3-form J, which includes both charge density and current density in spacetime.[18]

This formulation is not merely an aesthetic choice; it is a requirement for special relativity. The equations $dF = 0$ and $d * F = J$ are naturally Lorentz-covariant, meaning they take the same form in any inertial frame.[13] For the modern electrical engineer, this provides the correct foundation for understanding high-speed signal propagation, relativistic beam physics, and the operation of global positioning systems.[30]

Geometric Discretization and Computational Electromagnetics

A final, highly practical application of this framework is found in computational electromagnetics. The standard Yee scheme for Finite-Difference Time-Domain (FDTD) simulations is inherently a "differential forms" approach, even if it is often taught through vectors.[9]

The Yee Scheme as Discrete Forms

In the Yee grid, E and H are evaluated on the edges of the grid cells (where 1-forms are integrated), while D and B are evaluated on the faces (where 2-forms are integrated).[9] This "staggered" approach is not a numerical trick; it is a faithful representation of the geometry of Maxwell's equations. Because E is a 1-form, it *should* be evaluated on edges. Because B is a 2-form, it *should* be evaluated on faces.[9]

Using a framework called Discrete Exterior Calculus (DEC), researchers have proven that these staggered-grid methods automatically satisfy the conservation laws of electromagnetism (like $\nabla \cdot \mathbf{B} = 0$) because they preserve the discrete version of $d^2 = 0$.[9] For an EE student destined for a career in simulation and modeling, understanding differential forms provides the theoretical underpinning for why their software works—and how to recognize when it fails.[9]

Summary of the Geometric Perspective

The transition from vector calculus to differential forms can be summarized as a shift in "geometric imagination." Instead of arrows, the engineer visualizes the following [4]:

Physical Quantity	Geometric Object	Visual Interaction
Potential (Φ)	Points (0-form)	Evaluating height at a point.
Field Intensity (E, H)	Surfaces (1-form)	Piercing surfaces to measure work/effort.
Flux Density ($D, B, \cdot$)	Tubes (2-form)	Counting tubes passing through a surface.
Charge Density (ρ)	Cells (3-form)	Summing up boxes within a volume.
Exterior Derivative (d)	Boundary-finder	Linking cells to tubes and tubes to surfaces.
Wedge Product ($\wedge$)	Intersector	Crossing field surfaces to find energy or power.
Hodge Star ($*$)	Metric-converter	Rotating intensity surfaces into flux tubes.

Conclusions and Pedagogical Recommendations

The integration of Tristan Needham's visual differential geometry into the electrical engineering curriculum represents a significant leap in pedagogical efficacy. By grounding the abstract equations of Maxwell in the concrete geometry of surfaces and tubes, students move beyond rote memorization of vector identities toward a visceral understanding of the electromagnetic field.[1]

For the undergraduate peer, the practical takeaways are clear:

1. **Differentiate by Integration**: Always remember that the type of form is determined by what you integrate it over (lines, surfaces, or volumes).[4]

2. **Geometry is the Physics**: The topology of $dF = 0$ and $dG = J$ describes the "wiring" of the universe, while the metric of the Hodge star describes the local material environment.[19]
3. **Visualization is Calculation**: A clear mental picture of field surfaces and flux tubes often replaces pages of algebraic derivation, especially in complex coordinate systems.[1]

As electrical engineering continues to push into the regimes of nanophotonics, quantum computing, and high-frequency communication, the classical vector tools of the 19th century are becoming increasingly inadequate. The language of differential forms—geometric, intuitive, and relativistic—is the necessary upgrade for the 21st-century engineer.[1] By embracing the "drama" of geometry, the student not only learns to solve the equations but begins to understand the very nature of the space and time they are manipulating.[1]

Works cited

1. Visual Differential Geometry and Forms: A Mathematical Drama in Five Acts by Tristan Needham | Goodreads, accessed February 27, 2026, https://www.goodreads.com/book/show/54797008-visual-differential-geometry-and-forms
2. Visual Differential Geometry and Forms - Mathematical Association of America, accessed February 27, 2026, https://maa.org/book-reviews/visual-differential-geometry-and-forms/
3. Book Reviews A. Bultheel - KU Leuven, accessed February 27, 2026, https://people.cs.kuleuven.be/adhemar.bultheel/WWW/MISC/r118.php
4. Teaching electromagnetic field theory using differential forms - BYU ..., accessed February 27, 2026, https://scholarsarchive.byu.edu/cgi/viewcontent.cgi?article=1668&context=facpub
5. Teaching Electromagnetic Field Theory Using Differential Forms, accessed February 27, 2026, http://em.groups.et.byu.net/pdfs/publications/formsj.pdf
6. Visual differential geometry and forms - 1 · mihai.page, accessed February 27, 2026, https://mihai.page/vdgf-1/
7. the data of the original system and the tar- get system. The constructions are explicit, for constant coefficients, and they nee, accessed February 27, 2026, https://cdn.prod.website-files.com/5ee1777b5c6bb97683d92c6f/6673138bbf5a17c4ab515915_SIAM%20Review%20of%20VDGF.pdf
8. Maxwell's Equations in Differential Form, accessed February 27, 2026, https://ws.engr.illinois.edu/sitemanager/getfile.asp?id=195
9. Geometric Computational Electrodynamics with Variational Integrators and Discrete Differential Forms - Applied Geometry homepage, accessed February 27, 2026, https://geometry.caltech.edu/pubs/STDM14.pdf
10. Maxwell's Equations: Differential and Integral Forms - The Mathematical Wild, accessed February 27, 2026, https://mathwild.com/maxwells-equations-2-forms-2-meanings/
11. Visual Differential Geometry and Forms, accessed February 27, 2026,

https://api.pageplace.de/preview/DT0400.9780691219899_A40956557/preview-9780691219899_A40956557.pdf
12. Review of Tristan Needham's Visual Differential Geometry and Forms: A mathematical drama in five acts - Colorado State University, accessed February 27, 2026, https://www.math.colostate.edu/~clayton/research/papers/VDGF.pdf
13. Maxwell's Equations in Terms of Differential Forms - African Institute for Mathematical Sciences Library, accessed February 27, 2026, https://library.nexteinstein.org/wp-content/uploads/2022/04/solomon2009-10.pdf
14. Electromagnetics Made Easy: Differential Forms as a Teaching Tool, accessed February 27, 2026, https://em.groups.et.byu.net/pdfs/publications/fie.pdf
15. Differential Forms Crash Course, accessed February 27, 2026, https://math.okstate.edu/people/lebl/Differential_Forms_Crash_Course.pdf
16. Differential form - Wikipedia, accessed February 27, 2026, https://en.wikipedia.org/wiki/Differential_form
17. On the Visualization of Differential Forms, accessed February 27, 2026, https://math.uchicago.edu/~dmaia/documents/visualizing_diff_forms.pdf
18. Differential geometric formulation of Maxwell's equations, accessed February 27, 2026, http://home.lu.lv/~sd20008/papers/essays/Maxwell's%20equations.pdf
19. Maxwell's Equations, Hodge Theory, and Gravitation - arXiv, accessed February 27, 2026, https://arxiv.org/pdf/1305.6874
20. Geometric understanding of differential forms. - Math Stack Exchange, accessed February 27, 2026, https://math.stackexchange.com/questions/440816/geometric-understanding-of-differential-forms
21. Maxwell's Equations, accessed February 27, 2026, https://uspas.fnal.gov/materials/18ODU/2L%20Maxwell's_Equations.pdf
22. Introduction to differential 2-forms, accessed February 27, 2026, https://math.berkeley.edu/~wodzicki/H185.S11/podrecznik/2forms.pdf
23. differential forms and their application to maxwell's equations, accessed February 27, 2026, https://math.uchicago.edu/~may/REU2018/REUPapers/Eastman.pdf
24. Wedge Product -- from Wolfram MathWorld, accessed February 27, 2026, https://mathworld.wolfram.com/WedgeProduct.html
25. dg.differential geometry - Poynting vector and differential forms ..., accessed February 27, 2026, https://mathoverflow.net/questions/378504/poynting-vector-and-differential-forms
26. Lecture 2 Maxwell's Equations, Differential Operator Form - Purdue College of Engineering, accessed February 27, 2026, https://engineering.purdue.edu/wcchew/ece604s21/Lecture%20Notes/Lect2.pdf
27. Maxwell's equations and differential forms - MathOverflow, accessed February 27, 2026, https://mathoverflow.net/questions/72160/maxwells-equations-and-differential-forms
28. Poynting vector - Wikipedia, accessed February 27, 2026, https://en.wikipedia.org/wiki/Poynting_vector

29. A Plain Explanation of Maxwell's Equations - Fosco Connect, accessed February 27, 2026, https://www.fiberoptics4sale.com/blogs/electromagnetic-optics/a-plain-explanation-of-maxwells-equations
30. Maxwell's equations - Wikipedia, accessed February 27, 2026, https://en.wikipedia.org/wiki/Maxwell%27s_equations
31. How are the differential forms for Maxwell's Equations used? - Physics Stack Exchange, accessed February 27, 2026, https://physics.stackexchange.com/questions/466189/how-are-the-differential-forms-for-maxwells-equations-used